

Othman Shbeir

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Research Interests

Machine Learning; Artificial Intelligence; Computer Vision; Large Language Models; Natural Language Processing; Recommender Systems; Predictive Modeling; Applied AI in Education and Healthcare; Ethical and Human-Centered AI.

Education

Islamic University of Gaza, Gaza, Palestine

Bachelor of Science in Computer Science

Sep 2019 – Jul 2025

Rank: Ranked 2nd in Cohort: GPA: 95.34%

Thesis: HarmOni: Harmonizing Media and Social Experience

Academic Highlights:

- Designed and implemented an AI-powered multimedia synchronization and recommendation system integrating content-based filtering, SVD collaborative filtering, and popularity metrics.
- Completed advanced coursework in AI, ML, Computer Vision, Algorithms, Database Systems, Software Engineering, and Cloud Computing.
- Received multiple distinctions for academic excellence across all semesters.

Amideast Access Program

Dec 2017 – Aug 2019

English Language and Leadership Scholarship

Key Achievements:

- Champion, Climate Thinkathon (2023) for the leading team *ReUseIt* to first place with a sustainability-focused solution.
- Achieved C1-equivalent English proficiency and developed strong leadership skills.

Research Experience

Researcher — HarmOni (Undergraduate Thesis Project)

Sep 2022 – Jun 2023

Islamic University of Gaza

Research Focus: Multimedia synchronization, hybrid recommender systems, AI-driven user modeling

- Developed a hybrid recommendation engine combining content-based methods, SVD collaborative filtering, and popularity-based ranking.
- Implemented real-time synchronized media streaming using WebSockets and designed integrated social features.
- Applied LLMs to normalize filenames and enrich metadata using TMDB and MyAnimeList APIs.
- Built modular microservices using Flask, Docker, and NestJS; employed Agile development practices.

Researcher — Student Performance Prediction (Hackathon)

Jun 2023

Research Focus: Educational data mining, predictive modeling

- Built supervised learning models to identify at-risk students for early intervention.
- Designed preprocessing and evaluation pipelines using Pandas, NumPy, and scikit-learn.
- Produced interpretable metrics (accuracy, recall) and visual analytics dashboards.

Industry Experience

AI Engineer Intern — AURA for Integrated Solutions

Jun 2023 – Sep 2023

Applied AI and computer vision in production environments

- Built object detection and image recognition pipelines using YOLO, PyTorch, and TensorFlow/Keras.
- Contributed to 5+ large-scale AI solutions and collaborated with multidisciplinary teams.
- Supported model deployment, optimization, and integration with downstream systems.

Teaching Experience

Teaching Assistant — Programming 3 (Advanced Java)

Jul 2025 – Present

Islamic University of Gaza

Topics: JavaFX, Streams & Functional Programming, Spring Framework, Multithreading

- Led weekly laboratory sessions for 150+ students covering advanced object-oriented programming and concurrent computing.
- Conducted code reviews, problem-solving workshops, and individualized mentorship.
- Assisted in curriculum development, assignment design, grading, and instructional support.

Academic Projects

Bikeshare Data Analysis

Apr 2023

- Conducted exploratory data analysis to uncover seasonality, usage trends, and demographic patterns.
- Utilized Pandas, NumPy, and Matplotlib to produce interpretable analytics.

Sakila Database Analysis

May 2023

- Wrote optimized SQL queries for customer profiling and revenue insights.
- Built Power BI dashboards for data-driven business intelligence.

Honors & Awards

- **Second in Cohort, Computer Science** — Islamic University of Gaza 2025
- **Champion, Climate Thinkathon** — Amideast & VTech Road 2023
- **6th Place, Udacity Hackathon** — Student Grade Forecaster 2023
- **Access Alumni Leadership Recognition** — Amideast 2019
- **Academic Distinctions** — Islamic University of Gaza 2019–2025

Service & Community Engagement

Event Coordinator — Ranteesi Hospital Fun Day

Jun 2019

Organized recreational activities for children with cancer; coordinated logistics and volunteer teams.

IT Support Volunteer — Nasser Hospital

Jun 2019 – Jul 2019

Provided IT support, including hardware/software troubleshooting for hospital operations.

Operations Supervisor — Amideast Community Fun Day

Jul 2019

Managed volunteer teams and ensured smooth delivery of large-scale community events.

Technical Skills

Programming Languages: Python, Java, JavaScript, C, C++, SQL

Machine Learning: PyTorch, TensorFlow/Keras, YOLO, HuggingFace Transformers

Data Science: Pandas, NumPy, Matplotlib, Power BI, Tableau

Frameworks: Angular, React, Node.js, Nest.js, Spring, JavaFX

Tools: Git, Docker, Linux, L^AT_EX, AWS, Google Cloud

Languages

Arabic: Native proficiency

English: Professional working proficiency (C1)